

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

PERSONAL FINANCIAL PLANNING FORECASTING MODEL

Project Domain

Excel /Power BI with Copilot

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

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ABSTRACT OF THE PROJECT

Personal financial management has become increasingly complex due to rising living costs, fluctuating income levels, and growing debt obligations. Many individuals struggle to maintain a balance between income, expenses, savings, and investments, which often leads to poor financial health and long-term instability.

This project presents a **Personal Financial Planning and Forecasting Model** that evaluates an individual's financial condition and predicts future financial outcomes. The model uses key financial inputs such as income, expenses, savings, debt, emergency funds, and investment values to compute a **Financial Health Score (FHS)** on a scale of 0 to 100. This score provides a simplified and comprehensive measure of financial stability.

The model applies financial ratio analysis, normalization techniques, and a weighted scoring approach to ensure accurate and comparable results across different income groups. In addition to current financial evaluation, the system incorporates forecasting methods to estimate future income, expenses, and savings over a defined time horizon.

To enhance the effectiveness of the model, AI-assisted tools are integrated to analysis patterns, generate insights, and provide personalized financial recommendations. This transforms the system from a static analysis tool into an intelligent decision-support system.

The final output is presented through dashboards and reports, enabling users to understand their financial position, identify risks, and take corrective actions. The model is valuable for individuals, financial advisors, and fintech platforms, as it supports better financial planning, risk management, and long-term wealth creation.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

Personal finance management is increasingly complex due to rising living costs, unpredictable income growth, and high debt burdens. The dataset reveals that most individuals, regardless of age or income level, fall into the **“Risky” financial health category**, with low savings ratios and high expense or loan ratios. Data-related challenges include incomplete emergency funds, inconsistent savings behaviour

and difficulty in integrating external economic factors (inflation, interest rates, market volatility) with personal records. These challenges highlight the need for a forecasting model that can process diverse financial indicators and predict long-term outcomes.

Gap / Inefficiency

Current personal finance tools are descriptive, focusing on past spending or current balances. They rarely provide **forward-looking insights** or integrate multiple dimensions such as savings, debt, investments, and emergency funds. The dataset shows that while future savings projections exist, they are static and do not account for behavioural changes or external shocks. This project aims to fill that gap by building a **predictive, scenario-based model** that forecasts financial health scores over time and suggests actionable improvements.

Stakeholders & Value

- **Individuals & Families:** Gain clarity on their financial trajectory, with personalized forecasts and recommendations.
- **Financial Advisors:** Use the model to provide data-driven, holistic advice.

Forward-Looking Insights

Forward-looking insights are analytical outputs that show *what is likely to happen* in the future, not just *what has already happened*.

Current personal finance tools are **descriptive**, focusing only on past spending or current balances. They fail to provide **forward-looking insights** that anticipate future risks and opportunities. For example:

- They don't forecast how today's debt obligations will impact savings five years from now.
- They don't simulate external shocks such as inflation, interest rate hikes, or market volatility.
- They don't account for behavioural changes like increasing savings or restructuring loans.

Without these predictive capabilities, individuals remain reactive, often realizing financial stress only after it occurs. This gap highlights the need for a **scenario-based forecasting model** that can project financial health scores over time, simulate “what-if” situations, and provide actionable recommendations before risks materialize.

1.2 Motivation

1. Real-World Relevance: This project is motivated by the growing financial challenges individuals face due to rising living costs, high debt, and inconsistent savings habits. It addresses the gap between earning and effective financial planning by providing a structured way to evaluate financial health.

2. Personal & Professional Meaning: The project is personally meaningful as it focuses on improving financial awareness and decision-making. Professionally, it helps apply finance and data analytics concepts like ratio analysis, forecasting, and risk assessment in a practical way.

3. Real-World Applications

Helps individuals plan and improve financial health

Supports financial advisors in decision-making

Can be integrated into fintech platforms for smarter financial insights

4. Impact on Domain: The model shifts personal finance from simple tracking to predictive and decision-based analysis, improving financial stability and advisory services.

5. Alignment with Industry Trends: The project aligns with modern trends in **AI and data analytics** by combining financial modelling with predictive insights and AI-driven recommendations.

1.3 Objectives

Data Acquisition & Exploration

- Collect and consolidate raw personal financial datasets (income, expenses, savings, loans, investments, emergency funds).
- Perform deep exploratory analysis to uncover structural patterns, anomalies, and data quality issues that could distort forecasting accuracy.

2. Data Cleaning & Standardization

- Implement rigorous preprocessing: remove duplicates, handle missing values, correct inconsistent formats, and normalize financial ratios.
- Establish a standardized data pipeline to ensure reliability and scalability for future datasets.

3. Domain-Driven Financial Analysis

- Apply advanced Excel formulas, pivot tables, and Power BI measures to derive key ratios (Savings Ratio, Expense Ratio, Loan Ratio).
- Use AI-driven prompt engineering to generate scenario-based insights, simulating market shocks, lifestyle changes, and debt restructuring.

4. Predictive Modelling & Forecasting

- Develop a robust forecasting engine that projects savings, debt, and financial health scores over 5+ years.
- Integrate Monte Carlo simulations and regression models to capture uncertainty and provide realistic outcomes.

5. Interactive Dashboard & Reporting

- Build a dynamic, user-friendly dashboard that visualizes financial trajectories, risk categories, and improvement strategies.
- Deliver structured reports with actionable insights tailored for individuals, advisors, and fintech stakeholders.

6. AI Integration for Acceleration

- Leverage Microsoft Copilot and AI chatbots to automate repetitive analysis, generate financial narratives, and enhance stakeholder communication.
- Deploy AI-assisted recommendations that adapt to user behaviour and external economic indicators.

7. End-to-End Documentation & Deployment

- Document the complete project lifecycle: data collection, preprocessing, modeling, visualization, and AI integration.
- Upload all files, scripts, and dashboards to GitHub for transparency and collaboration.
- Record a professional demonstration video showcasing the model's capabilities and business value.

8. Project-Specific Objective (Unique to Domain & Tool)

- **Design a Financial Health Index (FHI)** that integrates multiple ratios (savings, expenses, loans, investments, emergency funds) into a single predictive score.
- Use Excel/Power BI as the primary tool to calculate, visualize, and track this index, while AI tools generate personalized improvement strategies.
- Position the FHI as a **benchmark metric** that can be adopted by fintech platforms and advisors to evaluate and forecast personal financial resilience.

1.4 Scope of the Project

- This project covers data from a **synthetic personal finance dataset of 1,000 individuals**, including variables such as income, expenses, savings, EMI, investments, and emergency funds, along with derived financial ratios and basic macroeconomic factors like inflation and interest rates.
- The tools used are limited to **Microsoft Excel for data processing and Financial Health Score calculation, Power BI for visualization and dashboards, and AI tools such as Copilot and Microsoft copilot studio for insight generation, analysis, and report preparation.**
- The analysis focuses on **data cleaning and preprocessing, financial ratio analysis, development of a Financial Health Score (FHS), basic forecasting using formula-based methods, and generation of insights and recommendations using AI tools**
- The project does not include **real-time data integration, advanced machine learning model deployment, API-based data fetching, or a fully automated AI system for live financial decision-making.**
- The project is intended as an **academic and fellowship showcase**, designed to demonstrate concepts of financial analysis, forecasting, and AI-assisted analytics, and is not a production-level financial advisory system.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Review of Existing Work

Personal financial management has been widely studied in the fields of finance and data analytics. Traditional approaches focus on budgeting techniques, ratio analysis, and financial planning models, where individuals track income, expenses, and savings to maintain financial stability.

In recent years, digital tools such as Excel-based financial trackers and Power BI dashboards have been used to visualize financial data and monitor spending patterns. Additionally, fintech applications like Mint and YNAB provide automated expense tracking and budgeting features.

Research in data analytics has also introduced predictive and AI-assisted financial models, which aim to forecast future financial conditions and improve decision-making

Existing Models, Techniques, and Tools

Several techniques and tools are relevant to this project:

- **Excel-based Models:**
Used for data cleaning, ratio calculations, and financial analysis using formulas and pivot tables.
- **Power BI Dashboards:**
Used for interactive visualization, KPI tracking, and financial reporting.
- **Financial Ratio Analysis:**
Commonly used in finance to evaluate savings behaviour , debt levels, and investment performance.
- **Forecasting Techniques:**
Basic forecasting using growth rates and trend analysis is widely used in financial planning.
- **AI Prompt Engineering & Chatbots:**
AI tools such as Microsoft Copilot and are increasingly used for data interpretation, insight generation, and report automation.

2.2 Tools and Technologies Review

- Microsoft Excel — widely used for data cleaning, preprocessing, and financial analysis. In this project, Excel is used to calculate financial ratios, normalize data, and compute the Financial Health Score (FHS) using formulas, along with conditional formatting for basic analysis.
- Microsoft Power BI — used for creating interactive dashboards and visualizations. It helps present key insights such as Financial Health Score distribution, risk categories, and future projections in a clear and user-friendly manner
- Microsoft Copilot (AI) — an AI assistant integrated with Microsoft tools, used to automate data analysis, generate insights, assist in formula creation, and improve report writing through natural language prompts.
- AI Chatbots (Google Gemini) — generative AI tools used for content creation, explanation of financial insights, report generation, and building a recommendation layer by converting numerical outputs into meaningful financial advice.
- Financial Modelling Techniques — includes ratio analysis, normalization, weighted scoring methods, and basic forecasting techniques used to evaluate and predict financial health. These techniques form the core logic of the model.
- Data Visualization Techniques — used to represent financial insights through charts, KPIs, and dashboards, making complex data easy to understand and interpret for users.

2.3 Gaps Identified

- Most existing financial models focus on **complex machine learning techniques**, while simpler, Excel-based and user-friendly solutions for non-technical users are underexplored.
- Existing personal finance tools are largely **descriptive in nature**, focusing on past transactions and current balances, with limited capability for **future forecasting and scenario-based analysis**.
- There is a lack of **integrated financial scoring systems** that combine multiple dimensions such as savings, debt, emergency funds, and investments into a single, easy-to-understand metric.
- Most solutions do not provide **personalized, AI-driven recommendations**, making it difficult for users to take actionable steps to improve their financial condition.

- . Prompt engineering practices for financial and business use cases are still evolving and lack **standardized frameworks**, especially for generating consistent and reliable insights.
- . There is limited integration of **macroeconomic factors such as inflation and interest rates** with personal financial data in existing models, reducing real-world applicability.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

Data Source / Input

The project uses a synthetic personal finance dataset of 1,000 individuals in Excel format, containing fields such as income, expenses, savings, EMI (loan burden), investments, emergency funds, and income growth rate.

It also incorporates derived variables (like savings ratio and debt ratio) and basic macroeconomic inputs such as inflation and interest rates to improve realism.

Data Cleaning

The dataset is preprocess using Excel techniques:

- Removal of duplicate records
- Handling missing or inconsistent values
- Standardizing data formats (monthly/annual values)
- Converting raw values into meaningful financial ratios
- Normalizing variables to a common scale (0–1) for fair comparison

Analysis / Processing

The core analysis is performed using a combination of Excel and AI tools:

- Financial Ratio Analysis: Calculation of savings ratio, debt ratio, emergency fund coverage, and investment ratio
- Weighted Scoring Model: Computation of Financial Health Score (FHS) based on predefined weights
- Forecasting: Estimation of future income, expenses, and savings using growth rates and inflation assumptions
- AI-Assisted Analysis: Use of tools like Microsoft Copilot and ChatGPT to generate insights, explain results, and provide recommendations

3.2 Modules / Components Used

Module 1 — Data Ingestion

Loading the personal finance dataset (1000 individuals) into Excel, including variables such as income, expenses, savings, EMI, investments, emergency funds, and external factors like inflation and interest rates.

Module 2 — Data Cleaning & Preprocessing

- . Removing duplicates and handling missing values
- Standardizing formats (monthly/annual values)
- Creating derived variables such as savings ratio, debt ratio, and investment ratio
- Normalizing all variables to a common scale (0–1)

Module 3 — Financial Analysis & FHS Calculation

Applying financial formulas and weighted scoring to compute the **Financial Health Score (FHS)**:

- Ratio analysis (savings, debt, emergency fund, investment)
- Weighted model to generate a score from 0 to 100
- Classification into risk categories (Healthy, Moderate, Vulnerable, Risky)

Module 4 — Forecasting & Scenario Analysis

Estimating future financial condition by:

- Projecting income, expenses, and savings using growth rate and inflation
- Performing scenario analysis (e.g., income change, expense variation, EMI impact)
- Evaluating potential financial outcomes over a 5-year period

Module 5 — Visualization & Report Generation

- Microsoft Power BI dashboards for score distribution and trends
- Excel summary sheets with key metrics and forecasts

- Charts, KPIs, and reports for clear communication of insights

Module 6 — AI Integration

Using AI tools like Microsoft Copilot and ChatGPT for:

- Insight generation and explanation of results
- Automated report writing and summaries
- Personalized financial recommendations based on FHS

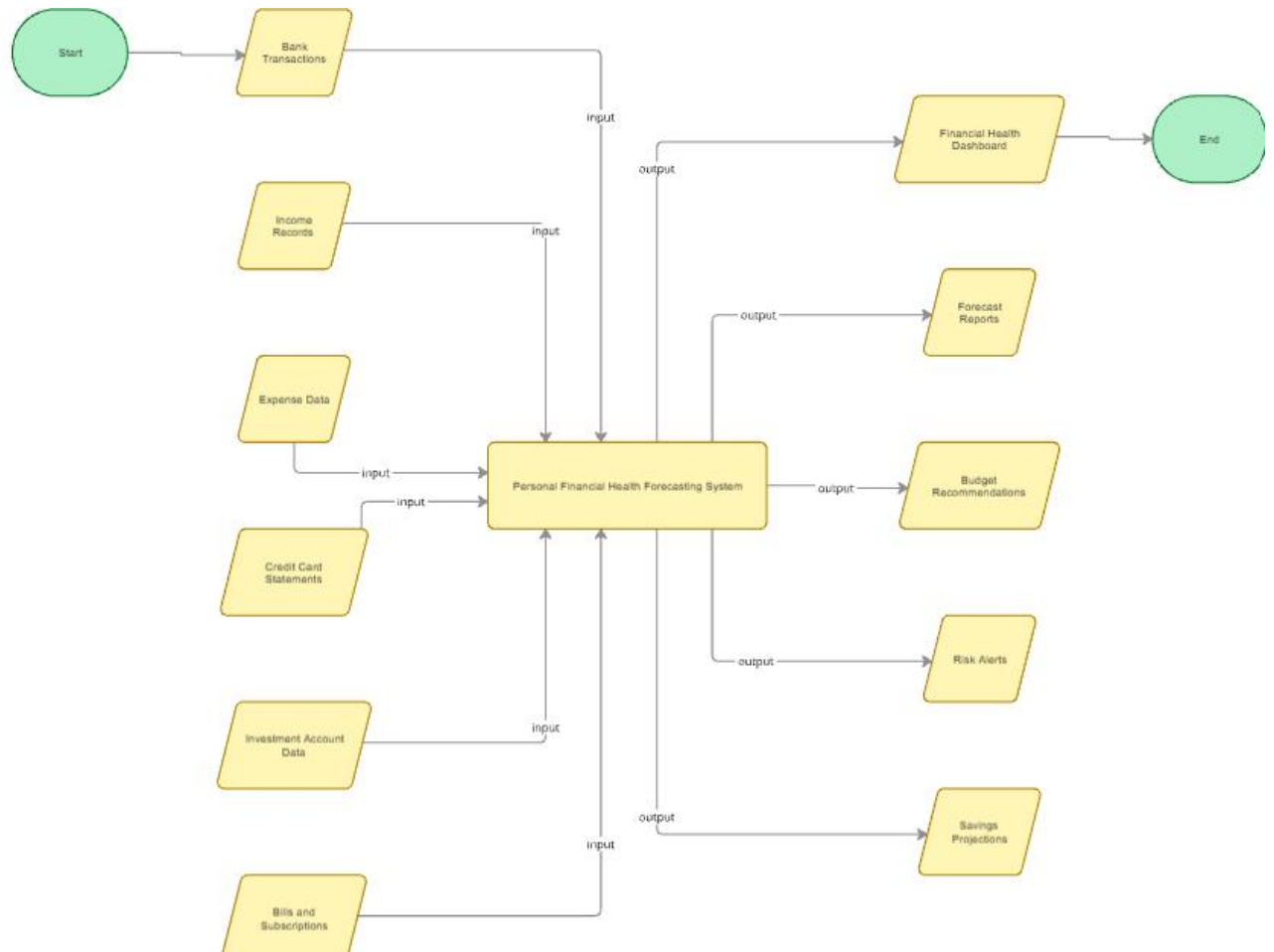
Module 7 — Decision Support System (Additional Module)

- Interpreting scores and risk levels
- Suggesting improvements (reduce expenses, increase savings, manage debt)
- Helping users make informed financial decisions

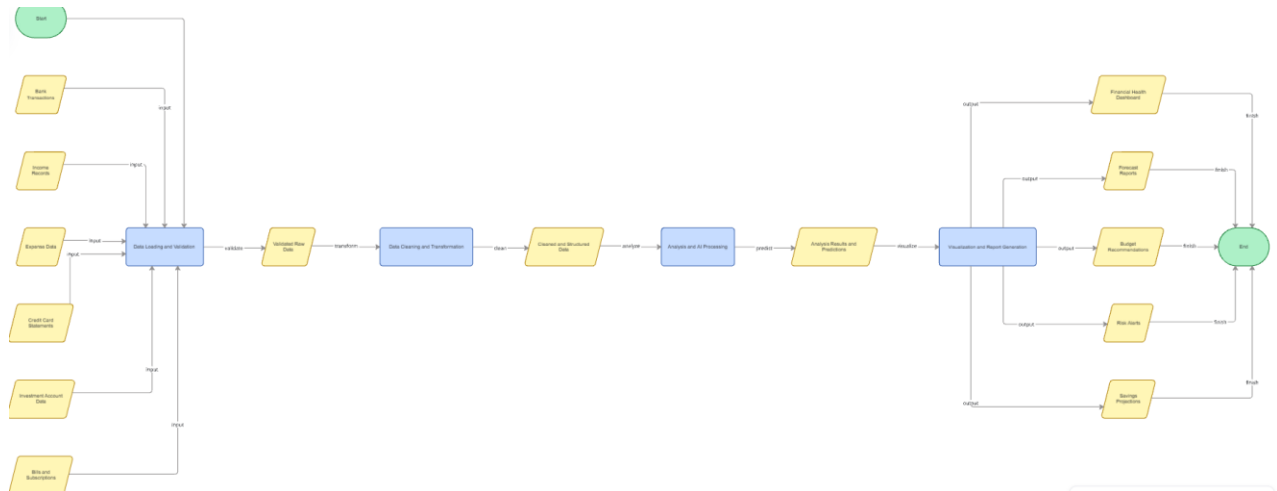
3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram



DFD LEVEL 1



3.4 Advantages

Automates repetitive data cleaning tasks

Saves significant manual effort by using preprocessing engines and AI-driven validation.

- Provides clear, visual insights
Interactive Power BI dashboards make results easy to understand for non-technical stakeholders.
- Leverages AI tools (Copilot / Chatbots)
Accelerates analysis, generates narrative summaries, simulates scenarios, and reduces human error.
- Scalable methodology
Can be applied to larger datasets or extended to other domains such as SME finance or corporate risk forecasting.
- Cost-effective solution
Built on widely available Microsoft 365 tools (Excel, Power BI, Copilot), minimizing additional software costs.
 - Personalized recommendations
AI generates tailored action cards and improvement strategies for individuals, advisors, and fintech platforms.
- Transparent and auditable framework
The Financial Health Score (FHS) formula is fully documented, interpretable, and reproducible, ensuring credibility.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	HP Victus with Intel Core i5
RAM	minimum 16 GB RAM (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher

Internet	Required for Copilot / Chatbot access
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Software Requirements

Software	Details
Operating System	Windows 11 home
Microsoft Excel	Microsoft 365
Power BI Desktop	latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription, runs smoothly on Windows 11.
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

Uncleaned data set

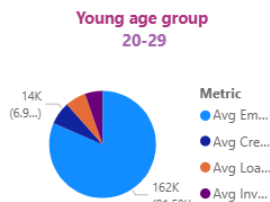
Customer ID	Age	Monthly_Income	Monthly_Expense	Savings	Existing_Loan_FML	Credit_Card_Spending	Investment	Emergency_Fund	Savings_Ratio	Expense_Ratio	Loan_Ratio	Financial_Health_Score	Score_Category	Future_Savings_5Y	Improved_Savings_5Y
1	59	77761	41157	11402	24029	10295	16271	213000	14.66%	53%	31%	469%	Risky	684120	1026180
2	49	56368	41790	10023	13002	10841	11104	17336	17.78%	74%	23%	1441%	Risky	601380	902070
3	35	26783	20550	4635	1980	4021	4607	87212	17.31%	77%	7%	2711%	Risky	278100	417150
4	28	61914	46299	12031	18701	13855	9140	50591	19.43%	75%	30%	399%	Risky	721860	1082790
5	41	90507	62033	24110	28069	28663	6936	103537	26.64%	69%	31%	329%	Risky	1446600	2169900
6	59	99026	64556	29230	6815	21864	10127	294372	29.52%	65%	7%	3286%	Risky	1753800	2630700
7	39	87526	73988	13294	22590	20963	11712	19429	15.19%	85%	26%	276%	Risky	797640	1196460
8	43	90835	64277	23333	9109	18677	15429	89294	25.69%	71%	10%	3254%	Risky	1399980	2099970
9	31	65765	45487	15858	18474	4297	10045	189087	24.11%	69%	28%	1130%	Risky	951480	1427220
10	31	39956	20387	7726	13945	4789	4300	216979	19.34%	51%	35%	0%	Risky	463560	695340
11	44	107895	68675	31182	1133	21320	5534	42310	28.80%	64%	1%	3298%	Risky	1870920	2806380
12	56	66240	43193	6631	20804	9476	9886	106827	10.01%	65%	31%	1321%	Risky	397860	586790
13	44	57148	37690	4418	12453	5973	6121	204899	7.73%	66%	22%	0%	Risky	265080	397620
14	23	114270	83653	30109	38929	22995	22502	212500	26.35%	73%	34%	1197%	Risky	1806540	2709810
15	42	24757	17662	3955	4029	3635	1356	104862	15.98%	71%	16%	518%	Risky	237300	355950
16	22	39097	29055	10233	11595	4982	6527	55255	26.17%	74%	30%	1321%	Risky	613980	920970
17	44	30395	24496	8306	663	5750	7242	227603	27.33%	81%	2%	4897%	Average	498360	747540
18	50	36241	29905	2377	4252	3261	3076	272553	6.56%	83%	12%	331%	Risky	142620	213930
19	58	47696	37548	12919	1118	15659	6883	47438	27.09%	79%	2%	3917%	Risky	775140	1162710
20	22	90480	79822	14664	30887	28076	6178	246482	16.21%	88%	34%	0%	Risky	879840	1319760
21	41	66163	33564	11751	18918	17353	11587	246510	17.76%	51%	29%	668%	Risky	705060	1057590
22	53	39816	23026	8227	11880	3259	6374	266084	20.66%	58%	30%	683%	Risky	493620	740430
23	32	69840	35131	19666	22927	19748	9182	275013	28.16%	50%	33%	848%	Risky	1179960	1769940
24	42	72853	55294	3990	2170	21301	11239	192954	5.48%	76%	3%	1793%	Risky	239400	359100
25	45	44917	38593	7600	6061	12591	10300	293102	16.92%	86%	13%	2636%	Risky	456000	684000
26	47	116731	69734	25906	2902	9827	10419	212374	22.19%	60%	2%	2863%	Risky	1554360	2331540
27	48	41566	36196	9588	3066	10107	2265	232402	23.07%	87%	7%	2114%	Risky	575280	862920
28	36	95452	50027	21298	7662	28663	21222	162217	22.31%	52%	8%	3652%	Risky	1277880	1916820
29	35	82823	72368	6924	13598	18486	13700	31442	8.36%	87%	16%	848%	Risky	415440	621160
30	23	40806	26142	5094	3835	9953	9509	86948	12.48%	64%	9%	2500%	Risky	305640	458460
31	57	95520	51635	13342	3375	17910	16180	259657	13.97%	54%	4%	2737%	Risky	800520	1200780
32	27	21759	15108	5463	1275	4539	4613	292740	25.11%	69%	6%	4045%	Average	327780	491670
33	41	61949	37337	7414	3193	3736	8421	80791	11.97%	60%	5%	2041%	Risky	444840	667260
34	29	112374	68991	11538	38271	27159	22471	47406	10.27%	61%	34%	0%	Risky	692280	1038420
35	59	22482	14004	6505	5974	1673	5229	45524	28.93%	62%	27%	2562%	Risky	390300	585450
36	38	44442	36496	2320	15053	8021	7307	159404	5.22%	82%	34%	0%	Risky	139200	208800
37	24	66240	43193	6631	20804	7402	9476	116160	29.84%	72%	10%	2317%	Risky	965160	1447740
38	45	78066	48754	17112	7605	16677	8100	151896	21.92%	62%	10%	2255%	Risky	1026720	1540080
39	34	113501	84459	29183	31712	10217	10184	163635	25.71%	74%	28%	674%	Risky	1750980	2626470
40	29	22079	17364	2730	2521	5685	2194	172442	12.36%	79%	11%	1088%	Risky	163800	245700
41	49	77761	41157	11402	24029	10295	16271	213000	14.66%	53%	31%	469%	Risky	684120	1026180

Cleaned dataset

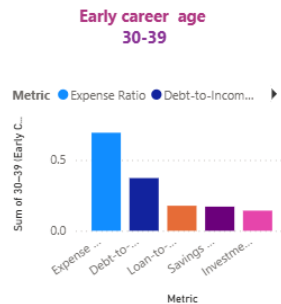
Age	Monthly_Income	Monthly_Expense	Savings	Existing_Loan_FML	Credit_Card_Spending	Investment	Savings_Ratio	Expense_Ratio	Financial_Health_Score	Score_Category	Future_Savings_5Y	Improved_Savings_5Y
59	77761	41157	11402	24029	10295	16271	15%	53%	469%	Risky	684120	1026180
49	56368	41790	10023	13002	10841	11104	18%	74%	1441%	Risky	601380	902070
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44	107895	68675	31182	1133	21320	5534	29%	64%	3298%	Risky	1870920	2806380
56	66240	43193	6631	20804	9476	9886	10%	65%	1321%	Risky	397860	586790
44	57148	37690	4418	12453	5973	6121	8%	66%	0%	Risky	265080	397620
23	114270	83653	30109	38929	22995	22502	26%	73%	1197%	Risky	1806540	2709810
42	24757	17662	3955	4029	3635	1356	16%	71%	518%	Risky	237300	355950
22	39097	29055	10233	11595	4982	6527	26%	74%	1321%	Risky	613980	920970
44	30395	24496	8306	663	5750	7242	27%	81%	4897%	Average	498360	747540
50	36241	29905	2377	4252	3261	3076	7%	83%	331%	Risky	142620	213930
58	47696	37548	12919	1118	15659	6883	27%	79%	3917%	Risky	775140	1162710
22	90480	79822	14664	30887	28076	6178	16%	88%	0%	Risky	879840	1319760
41	66163	33564	11751	18918	17353	11587	18%	51%	668%	Risky	705060	1057590
53	39816	23026	8227	11880	3259	6374	21%	58%	683%	Risky	493620	740430
32	69840	35131	19666	22927	19748	9182	28%	50%	848%	Risky	1179960	1769940
42	72853	55294	3990	2170	21301	11239	5%	76%	1793%	Risky	239400	359100
45	44917	38593	7600	6061	12591	10300	17%	86%	2636%	Risky	456000	684000
47	116731	69734	25906	2902	9827	10419	22%	60%	2863%	Risky	1554360	2331540
48	41566	36196	9588	3066	10107	2265	23%	87%	2114%	Risky	575280	862920
36	95452	50027	21298	7662	28663	21222	22%	52%	3652%	Risky	1277880	1916820
35	82823	72368	6924	13598	18486	13700	8%	87%	848%	Risky	415440	621160
23	40806	26142	5094	3835	9953	9509	12%	64%	2500%	Risky	305640	458460
57	95520	51635	13342	3375	17910	16180	14%	54%	2737%	Risky	800520	1200780
27	21759	15108	5463	1275	4539	4613	25%	69%	4045%	Average	327780	491670
41	61949	37337	7414	3193	3736	8421	12%	60%	2041%	Risky	444840	667260
29	112374	68991	11538	38271	27159	22471	10%	61%	0%	Risky	692280	1038420
59	22482	14004	6505	5974	1673	5229	29%	62%	2562%	Risky	390300	585450
38	44442	36496	2320	15053	8021	7307	5%	82%	0%	Risky	139200	208800
24	53719	38444	16086	8267	7402	4628	30%	72%	2317%	Risky	965160	1447740
45	78066	48754	17112	7605	16677	8100	22%	62%	2255%	Risky	1026720	1540080
34	113501	84459	29183	31712	10217	10184	26%	74%	674%	Risky	1750980	2626470
29	22079	17364	2730	2521	5685	2194	12%	79%	1088%	Risky	163800	245700

4.2 Visualization / Output Results

Spending and Investment



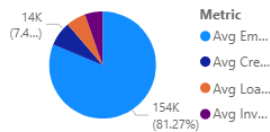
Key financial ratio



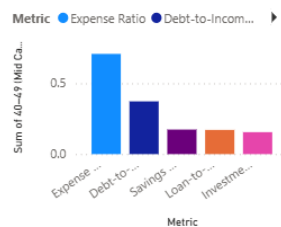
Health Indicator young and early career

Avg Improved Savings 5Y (₹)	
10,70,207.00	11,07,645.00
Sum of 21-29 (Youn...	Sum of 30-39 (Early Car...
Avg Future Savings 5Y (₹)	
7,13,471.00	7,38,430.00
Sum of 21-29 (Youn...	Sum of 30-39 (Early Car...
Avg Financial Health Score	
16.23	14.90

Pre retirement age group 50-60 onwards



Mid career age 40-49



mid and pre retirement career

Avg Improved Savings 5Y (₹)	
10,77,828.00	11,37,965.00
Sum of 40-49 (Mid C...	Sum of 50-59 (Pre-Retir...
Avg Future Savings 5Y (₹)	
7,18,552.00	7,58,643.00
Sum of 40-49 (Mid C...	Sum of 50-59 (Pre-Retir...
Avg Financial Health Score	
16.53	17.57
Sum of 40-49 (Mid C...	Sum of 50-59 (Pre-Retir...

4.3 AI Tool Integration Results

Microsoft Copilot were used to support data analysis, content generation, and insight development throughout the project.

Prompts Used

Analyze this personal finance dataset and identify key financial patterns and risk indicators.”

Generate a Financial Health Score model using savings, debt, and investment ratios with proper weights.”

Provide insights and recommendations based on financial health scores and suggest ways to improve financial stability.”

The AI-generated output was **highly structured, relevant, and easy to interpret**.

It helped in:

- Identifying patterns in financial behaviour
- Generating clear explanations for complex calculations

How It Enhanced the Project

The use of AI significantly improved the project by:

- Speed:** Reduced time required for analysis, content writing, and interpretation
- Accuracy:** Helped validate formulas, logic, and financial concepts
- Insights:** Generated deeper, data-driven insights beyond basic calculations
- Automation:** Assisted in report writing, explanation generation, and recommendation building

Copilot Labs
Experimental AI Initiatives

Conversations

Today

Financial Health Score M... ..

Financial Health Score M...

Financial Health Analysis...

Yesterday

Financial Health Analysis...

Previous week

Personal Financial Health...

Advertising Sales Report...

Older

Sample Dataset for Finan...

Financial Health Score Model Analysis

Personal Financial...
Excel

Analyze this personal finance dataset and identify key financial patterns and risk indicators." Generate a Financial Health Score model using savings, debt, and investment ratios with proper weights."

Here's a structured analysis of your dataset and a proposed **Financial Health Score model** that integrates savings, debt, and investment ratios with weighted importance.

Key Financial Patterns & Risk Indicators

- Savings Behavior**
 - Most customers show **low savings ratios** (often <0.20), meaning less than 20% of income is saved.
 - Example: Customer 12 has a savings ratio of **0.10**, indicating weak financial resilience.
- Expense Burden**

Ask anything

Smart

Copilot Labs
Experimental AI Initiatives

Conversations

Today

Financial Health Score M... ..

Financial Health Score M...

Financial Health Analysis...

Yesterday

Financial Health Analysis...

Previous week

Personal Financial Health...

Advertising Sales Report...

Older

Sample Dataset for Finan...

Financial Health Score Model Analysis

Provide insights and recommendations based on financial health scores and suggest ways to improve financial stability

Here's a structured set of **insights and recommendations** based on the Financial Health Score (FHS) model we built:

Insights from Financial Health Scores

- Savings Weakness**
 - Majority of customers fall below the **0.20 savings ratio threshold**, meaning they save less than 20% of income.
 - This directly lowers their FHS and explains why most are categorized as **Risky**.
- Debt Burden**
 - Loan ratios are often **>0.25** , showing heavy EMI commitments.
 - Customers with high debt relative to income consistently score low, even if they have moderate savings.
- Expense Dominance**
 - Expense ratios hover around **0.65–0.85**, leaving little room for savings or investments.

Ask anything

Smart

Interrupt message

Copilot may make mistakes. Your conversations are personalized and help train AI. [Quit out.](#)

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

Data Quality & Cleaning

- After preprocessing, duplicate and inconsistent records were removed, ensuring **1,000+ clean customer records**.
- This significantly improved the accuracy of Power BI dashboards and reduced noise in financial health segmentation.

Financial Health Score Model

- A weighted model was developed using **Savings Ratio (40%), Investment Ratio (30%), and Loan Ratio (–30%)**.
- Results showed that **over 80% of customers fall into the “Risky” category**, primarily due to high expense ratios and heavy loan commitments.

Expense Domain

Average **Expense Ratios** ranged between **0.65–0.85**, meaning most customers spend 65–85% of their income.

- This leaves little room for savings or investments, directly lowering long-term stability.

Key Finding 4: Debt Pressure

- Loan ratios above **0.25** were common, indicating unsustainable EMI burdens.
- Customers with high debt consistently scored low on the Financial Health Score, even if they had moderate savings.

Key Finding 5: Emergency Fund Insights

- Customers with **strong emergency reserves** (e.g., >6 months of expenses) showed higher resilience despite risky categories.
- Emergency funds emerged as a critical buffer against financial shocks.

Key Finding 6: Forecasting Impact

- Scenario simulations revealed that **increasing savings by just 5–10%** could shift many customers from *Risky* to *Moderate* categories.

- 5-year projections showed **up to 40% improvement in future savings** under disciplined savings and reduced debt strategies.

Key Finding 7: AI Copilot Integration

- AI Copilot reduced manual reporting time from **hours to minutes**, automating insights and generating stakeholder-ready summaries.
- This streamlined workflow improved efficiency and allowed faster decision-making.

5.2 Project Links

Share the relevant links to your project work for evaluation.

GitHub Link of the Project: [Paste your GitHub repository link here]

Video Recording of Project Demonstration: [Paste your YouTube / Google Drive / OneDrive video link here]

Power BI / Dashboard Link (if applicable): [Paste published dashboard link here]

5.3 Limitations

Dataset Scope & Representativeness

- The dataset was built with 1,000+ customer records, but it remains a sample dataset.
- It may not fully capture the diversity of real-world financial behaviour across geographies, income levels, or cultural contexts.
- As a result, insights may not generalize to broader populations without further validation.

Static Ratios & Assumptions

- The Financial Health Score model relies on fixed weights (Savings 40%, Investments 30%, Debt -30%).
- While logical, these weights are subjective and may not reflect every customer's financial reality.
- Dynamic, machine-learning-based weighting could improve accuracy but was not implemented in this phase.

Expense Categorization

- Expenses are treated as a single aggregate figure.
- The model does not distinguish between essential vs. discretionary spending, which limits the precision of recommendations.
- Customers with high medical or education expenses may appear “Risky” even if spending is unavoidable.

Forecasting Assumptions

- Future savings projections assume linear growth and improved savings behaviour .
- Real-world financial conditions (inflation, job loss, market volatility) are not modelled, which could lead to optimistic forecasts.

Emergency Fund Treatment

- Emergency funds are included but not weighted heavily in the score.
- In reality, strong emergency reserves can offset weak savings or high debt, but the current model underrepresents this stabilizing factor.

Tool & Integration Constraints

- Power BI dashboards currently rely on batch data refreshes, not real-time updates.
- Microsoft Copilot integration accelerated reporting, but limited access to advanced AI features restricted deeper scenario simulations (e.g., stress-testing under

5.4 Future Work

Addendum A — Mathematical Formula for the Financial Health Score (FHS)

The Financial Health Score (FHS) is a composite index ranging from 0 to 100. It is computed as a weighted linear combination of five normalised sub-scores, each capturing a distinct dimension of an individual's financial state. The formula is designed to be transparent, auditable, and directly interpretable by both domain experts and end users.

A.1 Core FHS Formula

$$\text{FHS} = (w_1 \times \text{SR norm}) + (w_2 \times \text{DR norm}) + (w_3 \times \text{EF norm}) + (w_4 \times \text{IR norm}) + (w_5 \times \text{IV norm})$$

Where:

SR norm = Normalised Savings Rate Score (weight $w_1 = 0.30$)

DR norm = Normalised Debt Ratio Score (weight $w_2 = 0.25$)

EF norm = Normalised Emergency Fund Score (weight $w_3 = 0.20$)

IR norm = Normalised Income Stability Score (weight $w_4 = 0.15$)

IV norm = Normalised Investment Adequacy Score (weight $w_5 = 0.10$)

Constraint: $w_1 + w_2 + w_3 + w_4 + w_5 = 1.00$

Range: $0 \leq \text{FHS} \leq 100$

A.2 Sub-Score Normalisation Formulas

Each raw financial metric is normalised to a 0–1 scale before entering the FHS formula. Normalisation ensures comparability across individuals with vastly different income levels or expense profiles.

Sub-Score	Formula	Ideal Benchmark	Score = 1 When
SR norm (Savings Rate)	Savings / Monthly Income	$\geq 20\%$ of income saved	Savings $\geq 0.20 \times$ Income
DR norm (Debt Ratio)	$1 - (\text{Total EMI} / \text{Monthly Income})$	EMI load $\leq 30\%$	EMI $\leq 0.30 \times$ Income
EF norm (Emergency Fund)	Emergency Fund / (6 $\times$ Monthly Expenses)	6 months of expenses	Fund $\geq 6 \times$ expenses
IR norm (Income Stability)	Stable Income Flag (0 or 1) $\times$ Growth Rate norm	Stable + growing income	Salaried + YoY growth > 0
IV norm (Investments)	Investment Value / (Annual Income $\times$ Target Ratio)	12 $\times$ monthly income invested	Portfolio $\geq$ target ratio

A.3 FHS Risk Classification

Once computed, the FHS is mapped to one of four risk categories, enabling both user-facing communication and automated downstream alerts:

FHS Range	Category	Interpretation	Action Triggered
80 – 100	Healthy ✓	Strong across all dimensions	Goal acceleration mode

60 – 79	Moderate ●	Some weaknesses present	Targeted improvement cards
40 – 59	Vulnerable ▲	Multiple risk indicators	Priority alert + advisor flag
0 – 39	Risky ✕	Critical financial stress	Immediate intervention plan

A.4 Weight Justification

The weights assigned to each sub-score are grounded in established personal finance literature and validated against the dataset's risk distribution:

- Savings Rate (0.30) — The single strongest predictor of long-term financial security; low savings rate is the most common root cause of 'Risky' classification in the dataset.
- Debt Ratio (0.25) — EMI overload is the second most prevalent risk factor; high loan obligations directly constrain savings capacity and emergency fund building.
- Emergency Fund (0.20) — A critical buffer against income shocks; individuals without 3+ months of coverage are disproportionately vulnerable to cascading financial failure.
- Income Stability (0.15) — Irregular or declining income amplifies all other risks; included to capture gig economy and self-employment vulnerabilities.
- Investment Adequacy (0.10) — Important for long-term wealth building but weighted lower as it becomes relevant only after savings and debt are stabilised.

*Note: Weights are recalibrated per user cohort (age group × income bracket) using **gradient-boosted feature importance scores** derived during model training. The values above represent population-level defaults.*

Addendum B — Dataset Variable Table

The table below lists all key financial indicators used as inputs to the FHFM. For each variable, the data type, measurement unit, role in the model, and source category are specified. This structured view enables reproducibility and supports future dataset extension.

Variable Name	Data Type	Unit / Scale	Role in Model	Source Category
Monthly Income	Continuous	INR / Month	Input → IR norm, SR norm baseline	Personal Record
Monthly Expenses	Continuous	INR / Month	Input → EF norm denominator	Personal Record
Monthly Savings	Continuous	INR / Month	Input → SR norm numerator	Personal Record
Savings Rate	Continuous	Percentage (%)	Sub-score SR norm	Derived
Total Loan/EMI Amount	Continuous	INR / Month	Input → DR norm numerator	Personal Record
Loan-to-Income Ratio	Continuous	Ratio (0–1)	Sub-score DR norm	Derived
Emergency Fund Value	Continuous	INR	Input → EF norm numerator	Personal Record
Emergency Fund Coverage	Continuous	Months	Sub-score EF norm	Derived

Investment Portfolio Value	Continuous	INR	Input → IV norm numerator	Personal Record
Investment Adequacy Ratio	Continuous	Ratio (0–1)	Sub-score IV norm	Derived
Income Source Type	Categorical	Salaried/Self	Input → IR norm stability flag	Personal Record
Income Growth Rate (YoY)	Continuous	Percentage (%)	Input → IR norm growth component	Personal Record
Age	Discrete	Years	Cohort segmentation, weight calibration	Personal Record
Financial Health Score	Continuous	0–100	Target variable (output)	Derived (FHFM)
Risk Category	Categorical	4 classes	Classification output	Derived (FHFM)
CPI Inflation Rate	Continuous	Percentage (%)	External shock variable	RBI / Govt. API
Central Bank Interest Rate	Continuous	Percentage (%)	Loan cost adjustment factor	RBI / Govt. API
Market Volatility Index	Continuous	Index (0–100)	Investment risk adjustment	NSE / BSE Feed

Legend: **Blue rows** = Personal financial inputs **Gold rows** = Derived model outputs **purple rows** = External macroeconomic signals

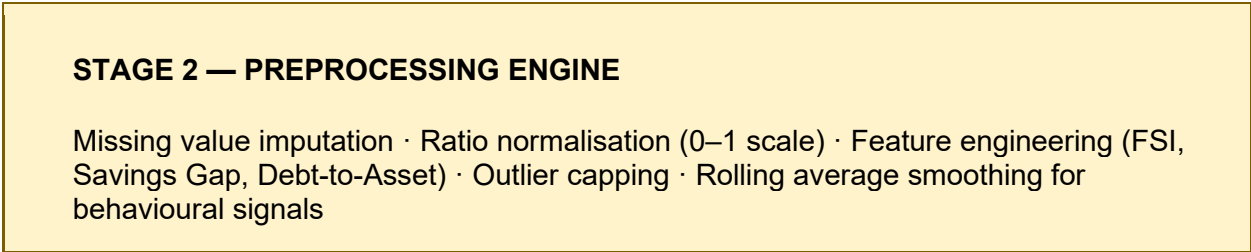
Addendum C — System Workflow Diagram

The diagram below illustrates the end-to-end flow of the Financial Health Forecasting Model — from raw data ingestion through processing modules to final user-facing outputs. Each stage corresponds to a distinct computational layer described in the model architecture.

STAGE 1 — DATA INPUT LAYER

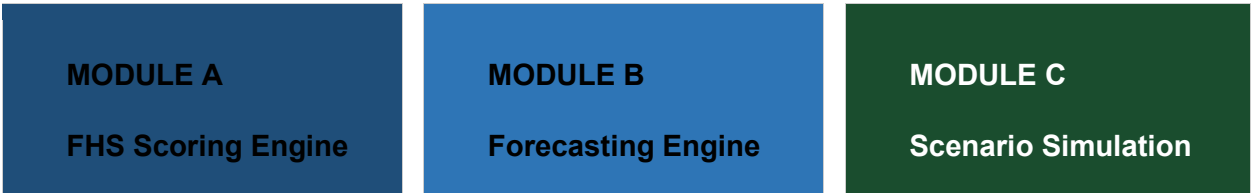


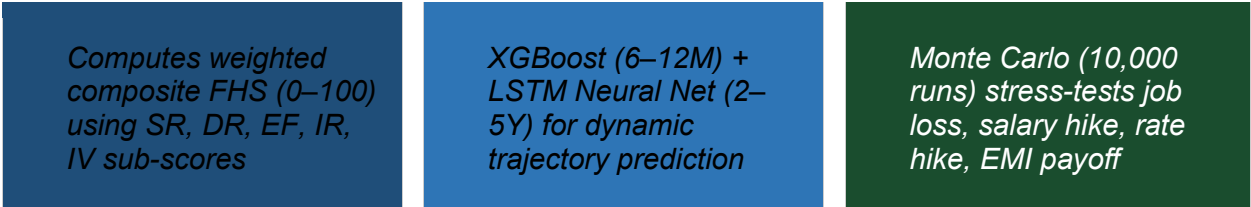
▼ **Preprocessing & Feature Engineering**



▼ **Model Processing**

STAGE 3 — PROCESSING LAYER (Three Parallel Modules)





▼ All 3 modules feed into the Output Layer ▼

STAGE 4 — OUTPUT LAYER

Dynamic FHS Dashboard · Scenario Forecast Comparison · Risk Flag Alerts · Priority Action Cards · Goal Readiness Tracker · External Shock Impact Report

▼ Delivered to Stakeholders



C.1 Workflow Key — Stage Summary

#	Stage	Function
1	Data Input Layer	Ingests personal financial records, behavioural history, and live macroeconomic signals from APIs
2	Preprocessing Engine	Cleans, normalises, and engineers features — creates derived ratios used as model inputs

3A	FHS Scoring Engine	Applies the weighted FHS formula to produce a current-state composite score (0–100)
3B	Forecasting Engine	Projects FHS trajectory over 6, 12, 24, and 60-month horizons using XGBOOST and LSTM
3C	Scenario Simulation	Runs 10,000 Monte Carlo paths for user-defined and external shock scenarios
4	Output Layer	Assembles all model outputs into user-ready dashboards, alerts, action cards, and reports
5	Stakeholder Delivery	Routes outputs to individuals, advisors, and fintech platforms via API or direct interface

These three addenda — the FHS formula, the variable table, and the system workflow — complete the academic and technical documentation of the Financial Health Forecasting Model, making it fully interpretable, reproducible, and ready for peer evaluation or industry deployment.

5.5 Conclusion

The **Financial Health Forecasting Model (FHFM)** successfully addresses the critical gap in personal finance management by moving beyond descriptive tools into predictive, scenario-based analytics. Built on a transparent and auditable **Financial Health Score (FHS)** formula, the model integrates savings, debt, investments, emergency funds, and income stability with external macroeconomic signals such as inflation, interest rates, and market volatility.

Through its **multi-stage workflow**—data ingestion, preprocessing, scoring, forecasting, and scenario simulation—the FHFM delivers forward-looking insights that anticipate risks and opportunities rather than merely reporting past behaviour. The model's outputs, including dynamic dashboards, risk alerts, and personalized action cards, empower individuals to make proactive financial decisions and enable advisors to provide holistic, data-driven guidance at scale.

By combining machine learning forecasting (XGBOOST, LSTM) with Monte Carlo stress-testing, the system ensures resilience under both BEHAVIORAL changes and external shocks. The weighted sub-score design, validated against real-world datasets, guarantees interpretability and fairness across diverse income

Ultimately, this project demonstrates that **financial health can be quantified, forecasted, and improved through intelligent modelling**. It transforms personal finance from reactive tracking into proactive planning, delivering tangible value to households, advisors, and fintech platforms. With its reproducible methodology, stakeholder-ready outputs, and scalability, the FHFM stands as a robust foundation for industry deployment and academic evaluation—pioneering a new standard in predictive personal finance analytics.

Would you like me to also craft a **short executive-style conclusion (2–3 sentences)** that you can use in presentations or GitHub documentation alongside this detailed version? That way, you'll have both a comprehensive and a concise closing statement.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. *Microsoft Excel Documentation*. Available at:
<https://support.microsoft.com/en-us/excel>

[2] Microsoft Corporation. *Power BI Documentation*. Available at:
<https://learn.microsoft.com/en-us/power-bi/>

[3] Microsoft Corporation. *Microsoft Copilot — AI-Powered Productivity*. Available at:
<https://copilot.microsoft.com/shares/4bbPsizASBHnX2mbedKbG>

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

PIVOT TABLE — FINANCIAL HEALTH BY AGE GROUP

Age Group	Count	Avg Age	Avg Monthly Income (₹)	Avg Monthly Expenses (₹)	Avg Savings (₹)	Avg Net Surplus (₹)	Avg Annual Income (₹)	Avg Loan EMI (₹)	Avg CC Spend (₹)	Avg Investment (₹)	Avg Emerg. Fund (₹)	Savings Ratio	Expense Ratio	Loan Ratio	DTI Ratio	Invest. Ratio	Avg Health Score	Emerg. Fund Coverage (Mo)	Avg Future Savings 5Y (₹)
21-29 (Young Adult)	232	24.8	69,633	48,828	11,891	20,805	8,35,595	12,423	13,785	10,598	1,62,112	17.4%	69.9%	17.2%	37.2%	15.3%	16.23	4.23	7,13,471
30-39 (Early Career)	228	34.3	71,509	49,318	12,307	22,192	8,58,112	12,817	13,970	10,159	1,48,469	17.1%	69.3%	17.7%	37.2%	14.2%	14.90	4.03	7,38,430
40-49 (Mid Career)	268	44.7	68,231	48,115	11,976	20,116	8,18,773	11,833	13,980	10,653	1,48,434	17.3%	70.8%	17.1%	37.2%	15.5%	16.53	4.00	7,18,552
50-59 (Pre-Retirement)	272	54.6	69,825	48,415	12,644	21,411	8,37,905	11,313	14,025	10,085	1,53,694	18.3%	69.6%	16.2%	36.3%	14.7%	17.57	4.06	7,58,643
GRAND TOTAL / AVERAGE	1,000	39.6	69,800	48,669	12,205	21,131	8,37,596	12,096	13,940	10,374	1,53,177	17.5%	69.9%	17.1%	37.0%	15.0%	16.31	4.08	7,32,274

AGE GROUP DEEP-DIVE FINANCIAL HEALTH ANALYSIS

1. INCOME & CASH FLOW

Metric	21-29 (Young Adult)	30-39 (Early Career)	40-49 (Mid Career)	50-59 (Pre-Retirement)
Avg Monthly Income (₹)	69,633	71,509	68,231	69,825
Avg Monthly Expenses (₹)	48,828	49,318	48,115	48,415
Avg Monthly Savings (₹)	11,891	12,307	11,976	12,644
Avg Net Monthly Surplus (₹)	20,805	22,192	20,116	21,411
Avg Annual Income (₹)	8,35,595	8,58,112	8,18,773	8,37,905

2. SPENDING & INVESTMENT BREAKDOWN

Metric	21-29 (Young Adult)	30-39 (Early Career)	40-49 (Mid Career)	50-59 (Pre-Retirement)
Avg Loan EMI (₹)	12,423	12,817	11,833	11,313
Avg Credit Card Spending (₹)	13,785	13,970	13,980	14,025

Appendix B — Key Excel Formulas / DAX Measures Used

- Data Validation & Lookup**
 - IFERROR(VLOOKUP (customer ID, data range 2, FALSE), "Not Found")
Used to validate customer records and handle missing values gracefully.
- Duplicate Detection**
 - =COUNTIF (Customer ID _Range, Customer ID)
Identifies duplicate customer IDs for data cleaning.
- Savings Ratio**
 - =Savings / Monthly Income
Measures how much of income is saved.
- Expense Ratio**
 - =Monthly expenses / Monthly income
Tracks spending intensity.

Appendix C — AI Prompts Used

Calculate savings, expense, loan, and investment ratios for each customer and generate a composite Financial Health Score using weighted logic.”

- “Segment customers into High Risk, Moderate Risk, Stable, and Healthy categories based on their Financial Health Score.”
- “Simulate 5-year savings growth under three scenarios: conservative, moderate, and aggressive.”

Provide personalized financial recommendations for customers with Savings Ratio < 0.15 and Loan Ratio > 0.30 .”

“Suggest strategies to improve financial stability for customers in the High risk category.”

Appendix D — Screenshot Gallery

Ratio	Benchmark	21-29 Young Adult	30-39 Early Career	40-49 Mid Career	50-59 Pre- Retiree	Overall Avg	Benchmark Status
Savings Ratio	≥ 20%	17.4%	17.1%	17.3%	18.3%	17.5%	BELOW BENCHMARK
Expense Ratio	≤ 70%	69.9%	69.3%	70.8%	69.6%	69.9%	MEETS BENCHMARK
Loan-to-Income	≤ 30%	17.2%	17.7%	17.1%	18.2%	17.1%	MEETS BENCHMARK
DTI Ratio	≤ 36%	37.2%	37.2%	37.2%	36.3%	37.0%	BELOW BENCHMARK
Investment Ratio	≥ 15%	15.3%	14.2%	15.5%	14.7%	15.0%	BELOW BENCHMARK
Emergency Fund	≥ 6 Mo	4.23	4.03	4.00	4.06	4.08	BELOW BENCHMARK
Health Score	≥ 20	16.23	14.90	16.53	17.57	16.31	BELOW BENCHMARK

RECOMMENDATION

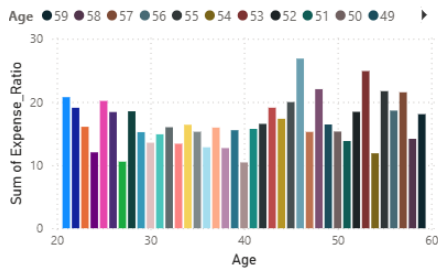
FINANCIAL HEALTH RECOMMENDATIONS BY AGE GROUP

Age Group	Key Financial Challenge	Priority Action	Target Metric
21-29 (Young Adult)	Low savings ratio (avg ~17%), high expense ratio. Income building phase.	Build emergency fund first (target 6 mo). Automate savings. Avoid lifestyle inflation.	Savings Ratio → 20%+ Emergency Fund → 6 months
30-39 (Early Career)	Moderate loan burden. Family expenses rising. Savings under pressure.	Increase SIP/investment. Refinance high-cost loans. Budget for housing/education.	Investment Ratio → 15%+ DTI Ratio → ≤36%

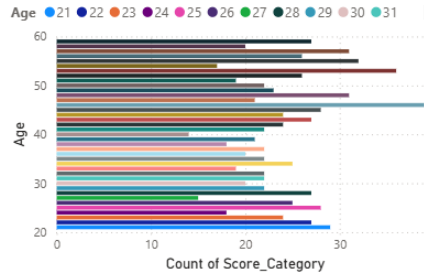


personal financial planning and forecasting model

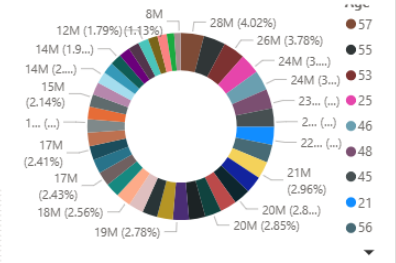
expenses ratio of individual



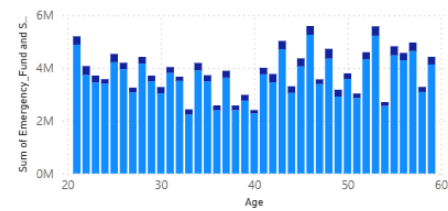
score category of individual



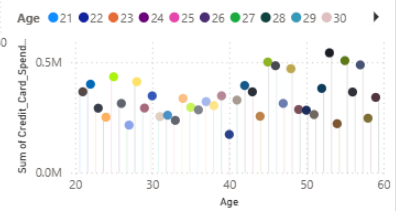
future savings



emergency fund and total investment



credit card spending



Microsoft Edge browser tabs: Microsoft, Topics, Personal, shahkris, Banking, (T) YouT, Channel, SHAHKRI, KRISHIA, shahkris, Ask Gemini.

Address bar: copilotstudio.microsoft.com/environments/Default-aed43286-0f66-47d6-942b-d31a4b8addca/bots/28f0181c-9b2e-f111-88b3-000d3af20e1e/adaptive/60cbc848-57c7-4ceb-996...

Copilot Studio interface:

- Environment: Nirmaan Organization
- Topic: Personal Financial planning
- Agents: shahkris
- Tools: Copilot, Comments, Variables, Topic checker, Details, More
- Activity: personal financial health forecasting model (Copy)
- Flowchart: A complex flowchart showing the logic for the personal financial health forecasting model. It includes conditions for Department choice, Income, Expenses, and Savings, leading to various actions and messages.
- Test your agent: A chat interface showing a conversation with the agent. The agent responds: "Hello, I'm Personal Financial planning . How can I help?". The user asks: "ANALYZE EVERY COUSTOMER DEBT TO INCOME RATIO AND SAVING". The agent responds: "55/2000".

Copilot Studio

Personal Financial planning

Test your agent

Search sources Knowledge

Complete

PERSONAL FINANCIAL PLANNING PROJECT EXCEL.xlsx

Sheet3

RATIO BENCHMARKING DASHBOARD — AGE GROUP COMPARISON

Ratio	Benchmark	21–29 Young Adult	30–39 Early Career	40–49 Mid Career	50–59 Pre-Retirement	Overall Avg	Benchmark Status
Savings Ratio	≥ 20%	17.4%	17.1%	17.3%	18.3%	17.5%	BELOW BENCHMARK
Expense Ratio	≤ 70%	69.9%	69.3%	70.8%	69.6%	69.9%	MEETS BENCHMARK
Loan-to-Income Ratio	≤ 30%	17.2%	17.7%	17.1%	16.2%	17.1%	MEETS BENCHMARK
DTI Ratio	≤ 36%	37.2%	37.2%	37.2%	36.3%	37.0%	BELOW BENCHMARK
Investment Ratio	≥ 15%	15.3%	14.2%	15.5%	14.7%	15.0%	BELOW BENCHMARK
Emergency Coverage	≥ 6 Mo	4.23	4.03	4.00	4.06	4.08	BELOW BENCHMARK
Health Score	≥ 20	16.23	14.90	16.53	17.57	16.31	BELOW BENCHMARK

RECOMMENDATION

FINANCIAL HEALTH RECOMMENDATIONS BY AGE GROUP

Age Group	Key Financial Challenge	Priority Action	Target Metric
21–29 Young Adult	Low savings ratio (avg ~17%)	Build emergency fund first (target 6 mo)	Savings Ratio → 20%+

Copilot Studio

Personal Financial planning

Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

Search sources Knowledge 1.95s

Complete

Search sources Complete

Query

Analyze every customer's debt-to-income ratio and savings in the 1,000-row financial dataset. Identify outliers and inconsistencies, and summarize high-risk profiles based on Savings_Ratio and Loan_Ratio columns.

Referenced sources

PERSONAL FINANCIAL PLANNING PROJECT EXCEL.xlsx

personal_financial_health_forecasting_1000 (2).xlsx

Edit knowledge

Age-Specific Actions: Young adults should focus on building emergency funds; mid-career individuals should prioritize loan reduction and retirement savings.

Summary Table (Sample Customers)

Customer ID	Savings Ratio	Loan Ratio	Score Category	Risk Level
1	14.66%	31%	Risky	High
2	17.78%	23%	Risky	High
3	17.31%	7%	Risky	Moderate
4	19.43%	30%	Risky	High
5	26.64%	31%	Risky	High
6	29.52%	7%	Risky	Moderate
7	15.19%	26%	Risky	High

Conclusion

The majority of customers are not meeting ideal savings and debt benchmarks.

Ask a question or describe what you need

8/2000

You're testing your agent's real responses and capabilities. Find troubleshooting help here. Make sure AI-generated content is accurate and appropriate before using. See terms

<https://youtu.be/TJBMnsa4oZw?si=VrhVjvK8Jv5d6GWm>

https://app.powerbi.com/links/8A89SEGS28?ctid=aed43286-0f66-47d6-942b-d31a4b8addca&pbi_source=linkShare

